2023 Eighth International Conference on Science Technology Engineering and Mathematics (ICONSTEM) | 979-8-3503-4779-1/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICONSTEM56934.2023.10142247

Credit Card Fraud Detection Using Machine Learning Based on Support Vector Machine

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Abstract: The usage of machine learning algorithm in the detection of fraudulent transactions is becoming more common. Most application systems, on the other hand, only catch fraudulent activityafter it has already taken place, rather than in real time. Detecting fraud is difficult because there are considerably fewererroneous transactions than legitimate ones. This data imbalance necessitates methods other than machine learning to handle it. Quantum machine learning (QML) has been used to develop detection frame work, which was then, implemented using SVMs supplemented with quantum annealing solvers. A total of twelve machine learning algorithms have been applied to testQML's detection capability, and their results have been compared to those of the QML application ontwo datasets: a non-time series of Israeli credit card transactions and a time series of Israeli bank loanapplications. The results reveal that, using the bank loan dataset, the quantum augmented SVM over takes the others in relationships of both speed and accuracy. The detection accuracy is comparable to those that use Israel credit card transaction data.By the detection time can be greatly improved for both data sets by using feature selection, although the increase in accuracy is minor. QML applications on time series data with significant imbalance have been shown to have promise, whereas standard machine learning methodologies have been shown to have worth when dealing withnon-time series data, as these results show. This research sheds light on how to choose the besttechnique for various datasets while keeping in mind the trade-offs between speed, accuracy, and price.

Key Words: Fraud detection, deep learning, machine learning,CCF.

I.INTRODUCTION

When someone else uses your credit card or account information to conduct an illegaltransaction, it's known as credit card fraud (CCF).Fraud may occur if a credit card is stolen, lost, orcounterfeited. Increased online buying has also led to a rise in "card-not-present" fraud, in which a criminal uses a victim's credit card data to completean online purchase. The development of e-banking in addition to other online payment atmospheres has led to an increase in fraud, such as CCF, causing in yearly losses of billions of dollars. One and only the most significant objective strendy today's world of digital payments is CCF detection.As a company owner, it is impossible to deny that cash less society is the future. If you want to grow your company, you can't rely on traditional payment methods like credit cards and debit cards. Customers may not always have money in their wallets when they come into the store. Debit and credit card payments are increasingly highly valued. Consequently, businesses will need to upgrade their systems so that they can accept a wide variety of payment options. This problem isonly going to become worse in the coming years[1].

From a total of almost 1.4 million identitytheft complaints in 2020, there were 393,207 CCF instances out of them. Only government papers andbenefits fraud have become more common forms of identity the ft inrecent years[2]. Fraud utilizing new credit card accounts is expected to reach 365,597 incidents in 2020 [3]. From 2019 to 2020, the amount of uniqueness the ft complaints has increased by 113%, with credit card uniqueness the ft information jumping by 44.6 percent [4]. $24.26 billion was spent on payment card fraud inthe worldwide economy in 2013. The USA is themost susceptible nation to credit theft, with 38.6percentof recordedlossesin2018.

Therefore, financial institutions should focus on implementing an automatic fraud detection scheme.

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Supervised CCF discovery aims to build a ML method founded on current transactional credit card expense data. Each transaction in the system is checked against a list of known frauds to determine whether or not the transaction is fraudulent. A number of basic issues are at play, together with the system's rapid response period, price sensitivity, and feature pre-processing. Machine learning is a branch of AI thatmakes predictions based on historical data patterns.[5]

There must been various researches that have employed ML models to tackle a variety of issues. Mobile cellular networks, computer networks, fraud detection in health care, medical malware detection, video surve illance detection, monitoring of position and location-based tracking of Android malware are just a few of the are as where deep learning algorithms have been put to use. ML algorithms may be used to detect creditcard frauds in the banking sector, and this study examines the practical use of ML in this area. When it comes to data categorization, the SVM is a supervised machine learning approach. As an example, it's used for picture acknowledgment [5], credit score calculation[5], and public safety monitoring [6]. While other classifier scan identify a hyper plane that divides the input data inthe support vector, SVM finds a hyper plane thatseparates the input data that is superior. Previous attempts at detecting credit card fraud relied on neural networks [6]. As a consequence, DL, a subfield of ML, is concentrating on DL methods atthe moment. Due to their promising results with a diversity of applications, such as computer visualization and natural language processing, deeplearning algorithms have recently gained a lot ofattention. A small minority, though, the use of deep neural nets in the detection of CCF has been studied.[7].

II. RELATEDWORK

There have been a number of studies donein the CCF detection sector. In this section, manystudies on CCF detection are presented. In addition, we would want to draw attention to research that suggests fraud detection as a solution to the issue of classim balance. Credit cards may be detected using a wide variety of methods. It's possible tocategorise the most relevant work in this field asDL, ML, CCF detection and ensemble ranking [8],[3], and user authentication techniques []. In orderto verify a credit card's authenticity, the technique shown in Figure1 is routinely employed. Passwords and biometrics are the two most common methods of securing an account. One may further categorise the use of biometrics to verifyidentity into three distinct categories: physiological, behavioral, and acombination of all three.

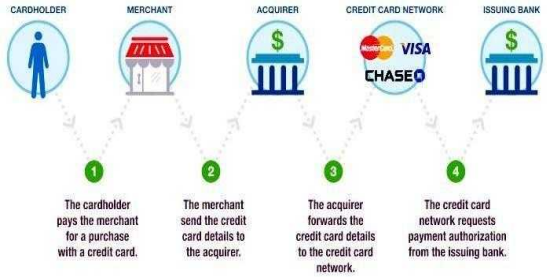

Fig1: The authorization procedure forcredit cards

Approaches to supervising machine learning (SMLE) Every branch of ML is capable of handling a different kind of learning assignment. Incontrast, ML learning has a variety of frameworks.CCF may be solved using ML techniques, like asrandom forests (RF). The random forest [3] is made out of the decision tree's many parts in random order. The RF method is used by the majority ofstudies. Use of (RF) and network analysis alongwith the model is an option for combining the twocomponents. This technique is known as APATE [10]. There are a variety of ML strategies available to researchers, including supervised learning and unsupervised learning. Avariety of machine learning methods, including LR, ANN, DT, SVMand NB, are routinely employed to identify CCF. Researchers may use these strategies to build reliable detection classifiers by combining them with ensemble techniques. An artificial neural network is a system of neurons and nodes linkedtogether. A contribution layer, a production layer, plus one or more hidden layers make up a feed-forward percept ronmulti layer. The first layer of input nodes is used to represent exploratory variables. All the hidden layer nodes are multiplied by a specified weight, and each of the input layers are multiplied by acertainbi as, and they are combined. The amount produced of each neuron for this summation is then transmitted to the next layer using an activation function. Finally, the output layer provides the algorithm's response. Random weights were utilised in the first set, and the training set was used in these cond. Back propagation [2], [6] were used to fine-tune theseweights. The Bayesian belief network is a visual representation of the interactions between collections of variables under

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uncertainty. Naïve Bayes' independence assumption was designed to allow for the possibility of correlations between variables.

III. Supervised Machine Learning Approaches

Nodes represent the quantities that may bechanged, while arcs connect the nodes that rely onone another. By linking the conditional probabilitytables for each node's variables, the possibilities foreachnode's variablearedependentonthenodevalues of its parent node [7, 8]. According to thefollowing, theBBN's computingsystemisasfollows: The initial stage is to identify a networkstructure, which may or may not be influenced bythe particular methods used to analyse thedata.AccordingtonaiveBayes,theconstant variablesare likewise discretized and purportedly distributednormallywhenthisnetworkstructureisfirstdeveloped.InBBN,eachnodeisanticipatedtoassumeitsmaternitiesinthenetwork[3],[9]andbeindependentofitsnoffspring.Markov'sconditionisrecognisedassuch.Supportvectormachines and regression issues are at the heart ofthe linear classification model. From [10] and [11],usingSVM,wecanidentifythemostrelevantpoints fromeach class. This is a kindof vectorknownasasup-portvector.Unsupervisedand

IV. Deep Learning Approach

CNN, DBN and DAE are examples of DLtechniques that are beneficial. [7], [5] have severallevels of data processing, illustration learning, and pattern categorization. The study of artificial neural networks liesattheheartofdeeplearning.Theconventionalmethoditisregardedasthebackpropagation model in terms of the scale of neuralnetworks [8],[6]. As the depth of neural networks increases, thebackpropagation algorithm'seffectiveness drops, which may lead to issues suchas alackoflocalobjectives andthe dilutionofmistakes.It'sagreataccomplishmentwhenadesign has a lot of complexity. Although they can't directly solve the optimization problem, the oretically they can [7], [8] do so within thetraining settings.

The deep beliefnetwork trainingmethod is frequently regardedasthemostsuccessfulexample of training in deep architecture. SVM, DT,and LR are among of the more traditional machinelearning techniques that have been suggested for CCF identification[3]. For hugedatasets,thesestandard techniques aren't particularly effective. DL methods likeas CNNs,whichareoftenusedinimageprocessing,haveastrongconnectiontothree-dimensional data. With the similar structurehidden layer and various quantity of channels ineach layer, the CNN is comparable to the ANN butwith additional convolution layers. Information thatmaybeused tocapturethemost importantinformationandautomaticallyreducethenumberof features is tied to the principle of shifting filtersvia word convolution, thus the CNN has become apopular tool for image-processing applications. Fortraining, the CNN does not need extensive data pre-processing.

V. System Architecture

Beforedeterminingwhetherornotthetime series data are stationary, a stationary test isconducted to determine if they're time series-basedor static.ADF and Kwiatkowski-Phillips-Schmidt-Shin are two commonstatisticaltestsusedtodeterminewhetheradatasetisstationary.Thisstudyutilisesbothmethods[2,3]toassess whether the time series data shown in Figure 2 isstationary.Thisresearchemploysthebasicdetrending procedures for non-stationary data: powertransformation, squareroot, and logarithmic transform. Finally, dimension reduction is used toeliminatethe "noise" aspectsofthe data.

WeuseLASSO [4] to construct more accurate prediction models by excluding variables that do not add to prediction accuracyorare"noises" that detract from it. Given a collection offeatures or attributes and the linear classifiers y, the shown in fig 2.

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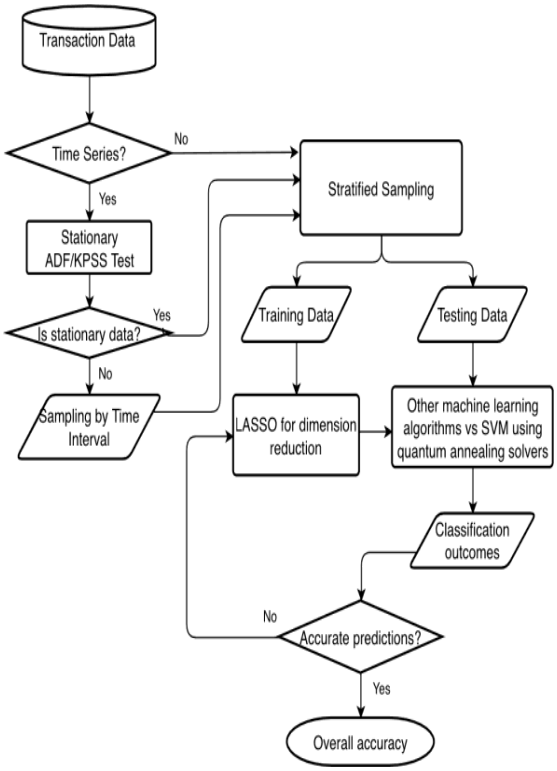

Fig 2: Flowchart of Transaction

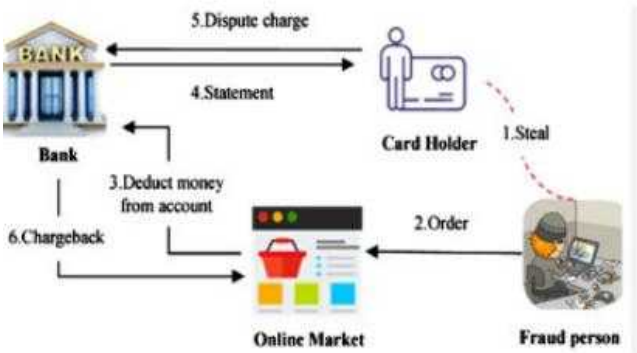

Fig 3: System Architecture

It's possible to reduce the value of the sparametersothatcertainindependentvariables(attributes)nolongercontributetothemodel,thereby improving fraud detection by reducing the number of irrelevantfeatures (vectors). SVM kernel functions will be presentexpressedasQUBO, in addition the kernel gatherings found by quantumannealing solverswillbeusedforextrapolativeanalysisoffrauddetectioninthemachinelearningtechnique.Atypicalmachinelearningalgorithmandtheq MLfakedetectionscheme's performancewillbecompared in fig 3.

VI .Evaluation Results

BymeansofnofeatureselectionorLASSOperformed,Tables3showhowSVM-QUBO comparesagainstother machinelearningtechniquesonthe LOANdataset. Thequicknessandoverallcorrectnessof SVM-QUBO aresuperior to all other machine learning algorithms, irrespectiveofwhetherfeatureselectionisused.Without feature selection, SVM-QUBO is 5 timesquicker than the fastest machine learning approach(Logistic Regression - balanced), 32 times fasterthanthe median, and 2813 times faster than the slowest method(Restricted BoltzmannMachine).

In comparison to the quickest machine learning (Linear Discriminant Analysis),SVM-QUBO is 3.8 times quicker, 16 times more rapidlythan the median and 2788 times earlier than the onethat takes the longest time to process (Restricted Boltzmann Machine).Comparedtothebest-performing outdated machine learning algorithm (Random Forest-balanced) without feature.

Table 1: Comparison of various algorithm

Method	Timein seconds	False Negative	False Positive	Correct Prediction	Overall Accuracy
SVM-QUBO	0.09263	760	52	10184	0.92615
Balanced Bagging	3.72162	742	146	10108	0.86413
Balanced RF	1.7605	331	3583	7082	0.63249
LDA	0.51514	752	34	10210	0.87088
LR	1.01368	763	0	10233	0.87218
LR- balanced	0.46915	347	4297	6352	0.5789
NN MLP	0.45193	763	0	10233	0.87231
RF	4.21333	761	3	10233	0.87231
RF-balanced	3.9576	763	1	10232	0.87243
Ensemble: RT-LR	3.85063	388	3238	7370	0.6456
COPOD	2.32763	710	1078	9208	0.79081
KNN	10.72006	720	950	9326	0.78926
RBM	260.5561	763	0	10233	0.87218

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This research demonstrates that QML isthequickestalgorithmavailable.FortheICCTdataset, which is fairly unbalanced and has a fraudrate of 23.8%, QML is not the ideal solution interms ofbothcost also performance shown in table 1.Thecorrectness of QML without feature selectionbepresentinferiortonineoutoftwelvemachinel earning algorithms, but it is greatly enhanced bymeans of feature selection (enhanced than 9 out of12).Byvariationsincorrectnessamong shown in fig 4 and 5.

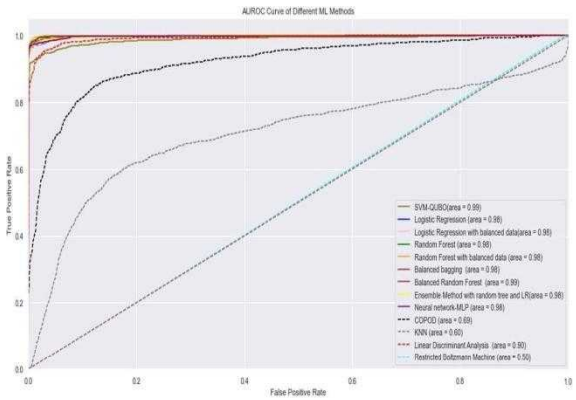

Fig 4: Figure3AUROCcurvescomparingSVM-QUBO

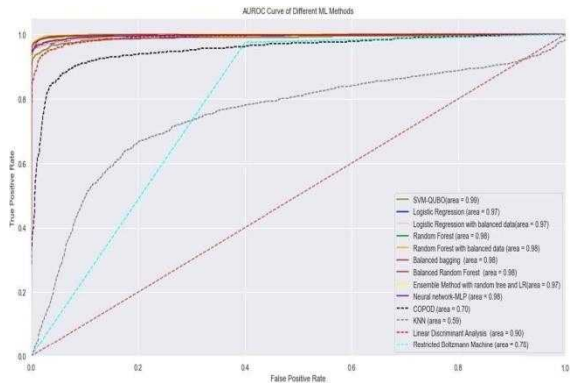

Fig 5: Figure3AUROCcurvescomparingSVM-QUBO with ICCTdataset

VII. CONCLUSION

Because to its processing powers, QML has gained increasing attention for its capacity to tackle complicated problems [9,20]. Due to difficulty of putting problems in to the QUBO format needed by quantum computing and the expense of such a technique, it is critical to identify the most important are as for QML application. S tudy compares the performance of a QML-based system for detecting fraud against that of twelve machine learning approaches. The results indicate the effectiveness of our suggested framework for detecting fraud and the better performance of QML, especially with a time-series-based, highly imbalanced, high-dimensional dataset. Our study contributes to the identification literature by out lining future QML research directions. Despite the increased interest in utilizing quantum to address real-world issues, our researchindicates that in order to give the ideal business solution, practitioners must consider the tradeoff between correctness, speed, and cost of computation, as well as the kind of data the system processes. Conventional machine learning algorithms are challenged by the processing of time-series, high-dimensional, and extremely imbalanced datasets. Quantum computing applications hold great potential for addressing these challenges. Conventional machine learning will continue to play a crucial role being the mostcost-effective and efficient approach for non-time-series data until quantum computing advances significantly.

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